Part Number	52-0164-AH	**Document No.**	WI-0523
Part Name	Actuator Housing, Aluminum Die Cast	**Revision**	A
Operation	Op 80: Deburr, Thread Chase, and Final Inspection	**Effective Date**	2026-10-05
Work Center	Finishing Bench 2	**Takt / Cycle Time**	6.0 min
Prepared By	Manufacturing Engineering	**Approved By**	Quality Manager

1. Purpose and Scope

This work instruction defines deburring, thread chasing, cleaning, and final dimensional inspection of the die cast aluminum actuator housing 52-0164-AH after machining at Op 70, releasing the part to the leak test station.

2. Safety Requirements and PPE

Requirement	Detail
PPE	Safety glasses, cut-resistant gloves (ANSI A3), hearing protection when air nozzle is in use.
Compressed air	Max 30 psi at the nozzle. Never direct air at yourself or others. Chip shield must be in place.
Hand tools	Deburr blades are sharp. Cut away from your body. Replace dull blades, do not force them.
Coolant residue	Parts may be oily. Wipe gloves and handles frequently to avoid drops.

3. Tools, Equipment, and Materials

Item	Description	Spec / Calibration
Deburring knife	Swivel-blade deburr tool, E-blade	Blade sharp, no chips
Thread chaser	M6 x 1.0 chase tap with tap handle	Inspect flutes each shift
Bore gage	Dial bore gage, 25 to 50 mm	Zeroed on master ring 32.000
Surface comparator	Ra comparison plate, N7	Visual reference
Air nozzle	OSHA-compliant, 30 psi max	Chip shield installed

4. Components / Materials Required

Component	Part / Spec No.	Qty
Housing, machined (from Op 70)	52-0164 (Op 70 complete)	1
Protective bore cap, 32 mm	C-3200-PE	1

5. Work Steps

Step	Instruction	Key Point / Specification
1	Pull one housing from the machining outbound cart. Confirm the Op 70 route stamp is present on the traveler.	Op 70 stamp present
2	Clamp the housing in the soft-jaw bench vise, main bore up. Snug only, do not distort the casting.	Soft jaws only
3	Deburr the parting line flash along both exterior ribs using the deburr knife.	No flash detectable by fingernail

Step	Instruction	Key Point / Specification
4	Deburr the main bore entry chamfer. One smooth pass, do not re-cut the chamfer angle.	Chamfer 0.5 x 45 deg preserved
5	Chase all four M6 x 1.0 mounting holes with the chase tap by hand. Back off a quarter turn for every half turn in.	Tap runs full depth by hand
6	Blow out all holes and the main bore with the air nozzle over the chip tray.	No visible chips or coolant
7	Wipe the datum face A with a clean blue towel.	Face clean and dry
8	Zero the bore gage on the 32.000 mm master ring at the bench.	Dial reads 0 on master
9	Measure the main bore at the top, middle, and bottom, in two orientations 90 degrees apart (6 readings).	32.000 to 32.025 mm, all readings
10	Record the max and min readings on the check sheet (or scan into the Digital Clipboard check sheet if the station tablet is active).	All readings in tolerance
11	Compare the bore surface finish to the N7 comparator plate under the bench light.	Ra 1.6 or better
12	Inspect the casting for porosity on machined faces. Any pore over 0.5 mm on a sealing face is a reject.	No pore > 0.5 mm on seal faces
13	Install the protective bore cap into the main bore.	Cap fully seated
14	Stamp the traveler Op 80 complete and place the housing in the leak test staging rack, max 12 per rack.	12 per rack max

6. Quality Checks

Characteristic	Specification	Method	Frequency
Main bore diameter	32.000 to 32.025 mm	Dial bore gage, 6 pts	Every part
Bore surface finish	Ra 1.6 max	Comparator plate	Every part
Thread condition	M6 chase tap full depth	Hand tap	Every part
Porosity, seal faces	No pore > 0.5 mm	Visual, 10x loupe if unsure	Every part
First piece + last piece	All above, logged	Full check	Shift start / end

7. Nonconforming Material

If any check fails or a defect is observed, STOP the operation. Tag the part with a red NONCONFORMING tag, record the part number, lot, and defect on the tag, and move the part to the quarantine rack. Notify the line lead and Quality immediately. Do not rework or scrap without an approved disposition.

8. Revision History

Rev	Date	Description	Approved
A	2026-10-05	Initial release	QM

Operator Signature: _____ Date: _____

Supervisor Verification: _____ Date: _____